

- Please read the questions very carefully before attempting to answer them.
- Write your answer on the answer sheet.
- You need to do 05 questions out of the following 06 questions.
- Use the table of Jumps provided with the question paper.
- Please be precise in your answers.

Question# 01: For the following code fragment, write the contents of registers where indicated. At the end write the updated values of the array.

```
[org 0x100]
    mov al, [num1] ;      al =
    mov bl, [num1 + 1] ;  bl =
    add al, bl          ;  al =
    mov bl, [num1 + 4] ;  bl =
    add al, bl          ;  al =
    mov [num1 + 5], al ;  al =
    mov ax, 0x4c00
    int 0x21
num1: dw 12ABh, 2D5Fh, 0ABCDh, 0
```

Question# 2: Show the values of the Carry, Zero, and Sign flags where indicated. Assume that all the flags are initially set.

```
mov al, 00001111b
SHL al, 2      ;    CF =      ZF =      SF =
XOR al, al     ;    CF =      ZF =      SF =
mov al, 6
cmp al, 5      ;    CF =      ZF =      SF =
mov al, 5
cmp al, 7      ;    CF =      ZF =      SF =
```

Question# 3: Let ax = 9, and bx = 20, write a code to perform a swap operation without using third register and memory operations?

Question# 4: Write a program to add ten numbers without a separate counter. *Do not use* quad word directive (Symbol for quad word declaration in 8088 in "QW" corresponds to 64 bit numbers).

Question# 5: Write the patch of the code for 16 bit multiplication:

multiplicand: dd 100

multiplier: dw 12

result: dd 0

start:

checkbit:

skip:

mov ax, 0x4c00

int 0x21

Question# 6: Fix the code for sorting the following array in ascending order and dry run it. Use the table of the jumps provided:

Org 0100h

jmp start

data: dw -60, 55, 55, 55

swap: dw 0

start: mov bx, 0

mov byte [swap], 0

loop1: mov dx, [data+bx]

cmp ax, [data-bx-2]

jbe noswap

mov dx, [data-bx-2]

mov [data-bx-2], ax

mov [data-bx], dx

mov byte [swap], 0

noswap: add bx, 4

cmp bx, 16

jne loop1

cmp byte [swap], 1

je start

mov ax, 0x4c00

int 0x21